



CSE623 Machine Learning: Theory and Practice

Topic:

Machine Learning Based Risk Parity Portfolio Construction for Financial Markets

Weekly Report 3

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1. Overview of Project:

Risk parity has emerged as a widely adopted portfolio construction framework that allocates capital by equalizing risk contributions across assets rather than maximizing expected returns. Popularized in institutional asset management as an “all weather” strategy, risk parity portfolios are designed to deliver stable performance across different market regimes. However, the effectiveness of risk parity critically depends on accurate estimation of asset volatilities and correlations.

Recent advances in machine learning and statistical learning theory provide tools for denoising and dynamically estimating covariance structures. Techniques such as shrinkage estimators, random matrix theory based eigenvalue filtering, and hierarchical clustering provide more robust representations of dependency structures (tail risks) for a multi-asset portfolio.

This project builds a risk parity portfolio using machine learning techniques. Using daily US financial assets data from 2015–2025, we construct long-only, fully invested portfolios. We then evaluate their out-of-sample performance from the year 2020 till 2025 and compare it with a benchmark.

2. Overview of Week 3:

Tasks accomplished this week:

- The return series was cleaned by removing missing values introduced during the return computation step. The first observation in each series, which lacks a prior price for differencing, was dropped and the index was reset to produce a clean dataset for covariance estimation.
- Multiple covariance estimation methods were implemented and organized into a modular class structure. These include the sample covariance matrix, constant correlation shrinkage, Ledoit–Wolf optimal shrinkage, and Oracle Approximating Shrinkage (OAS). This modular design allows seamless switching between risk models during portfolio construction.
- Eigen decomposition was performed on the estimated covariance matrices to extract principal components. Eigenvalues were sorted in descending order, cumulative explained variance was computed, and two filtering techniques were applied: an explained variance threshold and a Random Matrix Theory (Marčenko–Pastur) filter to separate signal from noise.
- A PCA-based risk parity portfolio was constructed by assigning equal risk to each retained principal component, converting factor-space weights back into asset-space weights, and normalizing them to sum to unity.
- A rolling window back-testing framework was built using a five-year (60-month) estimation window. For each window, the full pipeline—covariance estimation, PCA decomposition, eigenvalue filtering, and weight computation—was executed, producing a time series of portfolio weights across multiple method combinations.

3. Work Completed:

Data Cleaning and Preparation

- When computing asset returns from price data, the first observation in each series becomes missing because there is no prior price available for differencing. These missing values were removed and

the DataFrame index was reset, producing a clean returns dataset suitable for covariance estimation and downstream analysis.

Covariance Matrix Estimation

- The covariance matrix of asset returns is a critical input for risk parity portfolio construction. Because the sample covariance matrix is known to suffer from estimation noise—especially when the number of assets is large relative to the sample size—multiple estimation techniques were implemented to improve robustness.
- The historical sample covariance matrix was computed as the baseline estimator. It measures pairwise co-movement between assets, with variances on the diagonal and covariances on the off-diagonal elements.
- A constant correlation shrinkage estimator was implemented. This method constructs a structured target matrix that assumes all pairwise correlations are equal to the average sample correlation, and then forms a weighted combination of the sample covariance and the target. This reduces instability caused by noisy off-diagonal estimates.
- The Ledoit–Wolf shrinkage procedure was applied to compute the optimal shrinkage intensity automatically. Rather than choosing the blending weight manually, the algorithm estimates the noise level in the sample covariance and determines the optimal weight to assign to the shrinkage target. This produces a covariance estimator that reduces estimation error and improves out-of-sample portfolio performance.
- Oracle Approximating Shrinkage (OAS) was also implemented. OAS is particularly effective when the asset universe is large and the sample size is relatively small, as it automatically calibrates a shrinkage parameter to yield a more stable covariance estimate.
- All covariance estimation methods were organized into a unified class. This modular architecture allows the program to switch seamlessly between the sample covariance, Ledoit–Wolf shrinkage, OAS, and custom shrinkage estimators during portfolio construction and back-testing.

Principal Component Analysis and Eigenvalue Filtering

- Eigen decomposition was performed on each estimated covariance matrix. Mathematically, the decomposition solves $\Sigma v = \lambda v$, where λ denotes the eigenvalues and v the corresponding eigenvectors. Eigenvectors represent principal components (latent risk factors), while eigenvalues represent the variance explained by each factor.
- Eigenvalues were sorted in descending order so that the largest values—corresponding to dominant risk factors such as the market factor, sector factor, and style factor—appear first. Cumulative explained variance was computed to determine how many components are needed to capture most of the system’s risk.
- Two eigenvalue filtering methods were implemented. The first retains only principal components whose cumulative explained variance exceeds a specified threshold (for example, 80 percent). Components below the threshold are discarded as noise. The second method uses Random Matrix Theory: the Marčenko–Pastur distribution provides a theoretical boundary separating random noise eigenvalues from information-bearing eigenvalues. Eigenvalues falling below this boundary are treated as noise and removed, improving portfolio stability.

- Using the filtered eigenvectors, orthogonal risk factors were constructed. Each factor represents an independent source of risk in the market, such as overall market movement, sector rotation, or cross-asset volatility. By construction, these factors are uncorrelated.

PCA-Based Risk Parity Weight Construction

- Instead of allocating equal weight to individual assets, the portfolio was constructed so that each retained principal component contributes equal risk. Since each component has variance equal to its eigenvalue, the factor-space weights were scaled by the inverse square root of the eigenvalues, thereby equalizing risk contributions across factors.
- The factor-space weights were then mapped back into the original asset space using the eigenvector matrix. The resulting vector provides the portfolio weight for each asset. These weights were normalized so that they sum to one, ensuring a fully invested, long-only portfolio.
- The closed-form solution for PCA Risk Parity portfolio weights is:

$$w = V [1/\sqrt{\lambda_1}, \dots, 1/\sqrt{\lambda_k}]^T / (1^T V [1/\sqrt{\lambda_1}, \dots, 1/\sqrt{\lambda_k}]^T)$$

- This construction equalizes risk across orthogonal risk factors (principal components), avoids concentration in the dominant market mode (PC1), and produces diversified exposures to latent factors. Thus, PCA Risk Parity controls systematic factor risk rather than merely balancing individual asset volatility.

Rolling Window Portfolio Allocation Framework

- To simulate realistic portfolio management, the dataset was divided into rolling time windows of five years (60 months). For example, the first window uses data from January 2015 to December 2019, the second from February 2015 to January 2020, and so on. Each window provides the historical returns used to estimate the covariance matrix for that period.
- For each rolling window, the entire pipeline was executed: return data was selected, the covariance matrix was estimated, PCA was performed, eigenvalues were filtered, and PCA risk parity weights were computed. This produces a time series of portfolio weights, one vector per rebalancing date.
- The process was repeated using different combinations of covariance estimators and eigenvalue filtering methods. Variants include: sample covariance with explained variance filtering, sample covariance with random matrix filtering, Ledoit–Wolf covariance with explained variance filtering, and OAS covariance with explained variance filtering. This allows systematic comparison of how different risk estimation techniques influence portfolio allocation.
- The resulting portfolio weights for every rebalancing period were stored in a structured format indexed by date and asset name. This facilitates downstream performance evaluation and turnover analysis.

4. Challenges Encountered:

- The PCA Risk Parity construction, when applied with a simple budget normalization (weights summing to one), produced portfolios with both positive and negative weights. In a long-only setting, the resulting portfolio generated negative returns across several rolling windows. This indicated that imposing a long-only constraint on inherently long-short factor exposures distorts

the risk parity interpretation. To address this, we shifted to a long-short portfolio framework, where the optimization allows negative positions. However, implementing long-short risk parity requires solving a non-linear optimization problem, which does not always converge reliably. We referred to Roncalli and Weisang (2012), “Risk Parity Portfolios with Risk Factors,” which provides a rigorous treatment of risk budgeting in factor space and discusses conditions under which the optimization is well-posed.

- Deciding which covariance estimation method to pair with which eigenvalue filtering technique was a non-trivial design choice. Two filtering approaches were considered: the Marčenko–Pastur random matrix theory filter, which removes eigenvalues below a theoretically derived noise threshold, and the explained variance filter, which retains components explaining at least 80 percent of total variance. Each combination of covariance estimator (sample, Ledoit–Wolf, OAS, constant shrinkage) and filtering method produces a different set of retained factors, leading to meaningfully different portfolio allocations.
- The portfolio weights produced by the PCA Risk Parity procedure exhibited extreme magnitudes. In some rolling windows, individual asset weights reached as high as +9,900 percent (long) and as low as −6,600 percent (short). This results in extremely leveraged portfolios that are not practically feasible. The issue arises because the eigenvector matrix V can map moderate factor-space weights into very large asset-space weights when eigenvectors have elements of mixed sign and similar magnitude. Addressing this requires either explicit leverage constraints or a regularization procedure on the asset-space weights, which is a key focus for the next phase.

5. Plan for Week 4:

- Define a rebalancing threshold to determine whether portfolio weights should be updated at each rolling window or held unchanged. This threshold will be calibrated based on weight deviation between consecutive periods.
- Compute out-of-sample portfolio returns using the weight vectors generated from the rolling window framework and evaluate performance metrics including cumulative returns, annualized volatility, Sharpe ratio, and maximum drawdown.
- Compare the performance of all portfolio variants against a benchmark to assess the value added by PCA-based risk parity and different covariance estimators.
- Build a benchmark that matches the blend of asset classes present in our portfolio, ensuring a fair comparison.

References:

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